The number of points, at which the function

$f(x) = \max\{6x,\ 2+3x^2\} + |x-1|\cos\left|x^2 - \frac{1}{4}\right|,\ x \in (-\pi, \pi)$ , is not differentiable, is

_____.

**Response Type :** Numeric
**Evaluation Required For SA :** Yes
**Show Word Count :** Yes
**Answers Type :** Equal
**Text Areas :** PlainText
**Possible Answers :**
1

# Physics Section A

| | |
|---|---|
| **Section Id :** | 69527821 |
| **Section Number :** | 3 |
| **Section type :** | Online |
| **Mandatory or Optional :** | Mandatory |
| **Number of Questions :** | 20 |
| **Number of Questions to be attempted :** | 20 |
| **Section Marks :** | 80 |
| **Maximum Instruction Time :** | 0 |
| **Sub-Section Number :** | 1 |
| **Sub-Section Id :** | 69527821 |
| **Question Shuffling Allowed :** | Yes |
| **Is Section Default? :** | No |

**Question Number : 26 Question Id : 695278251 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

In a screw gauge when the circular scale is given five complete rotations it moves linearly by 2.5 mm. If the circular scale has 100 divisions, the least count of screw gauge is _____ mm.

**Options :**

695278851. $1 \times 10^{-2}$

695278852. $1 \times 10^{-3}$

695278853. $5 \times 10^{-2}$

695278854. $5 \times 10^{-3}$

**Question Number : 27 Question Id : 695278252 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

The increase in the pressure required to decrease the volume ($\Delta V$) of water is $6.3 \times 10^7$ N/m$^2$. The percentage decrease in the volume is _____.
(Bulk modulus of water = $2.1 \times 10^9$ N/m$^2$.)

**Options :**

695278855. 2 %

695278856. 3 %

695278857. 6 %

695278858. 4 %

**Question Number : 28 Question Id : 695278253 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

The time taken by a block of mass $m$ to slide down from the highest point to the lowest point on a rough inclined plane is 50 % more compared to the time taken by the same block on identical inclined smooth plane. Both inclined planes are at 45° with the horizontal. The coefficient of kinetic friction between the rough inclined surface and block is _____.

**Options :**

695278859. 3/4

695278860. 2/3

695278861. 5/9

695278862. 4/9

**Question Number : 29 Question Id : 695278254 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

Two nuclei of mass number 3 combine with another nucleus of mass number 4 to yield a nucleus of mass number 10. If the binding energy per nucleon for the mass numbers 3, 4 and 10 are 5.6 MeV, 7.4 MeV and 6.1 MeV, respectively, then in the process, $\Delta Mc^2$ = _____MeV.

**Options :**

695278863. 6.9

695278864. 7.9

695278865. 2.2

695278866. 4.3

**Question Number : 30 Question Id : 695278255 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

A solid sphere of mass $M$ and radius $R$ is divided into two unequal parts. The smaller part having mass $M/8$ is converted into a sphere of radius $r$ and the larger part is converted into a circular disc of thickness $t$ and radius $2R$. If $I_1$ is moment of inertia of a sphere having radius $r$ about an axis through its centre and $I_2$ is the moment of inertia of a disc about its diameter, the ratio of their moment of inertia $I_2/I_1$ = _____.

**Options :**

695278867. 35

695278868. 70

695278869. 140

695278870. 210

**Question Number : 31 Question Id : 695278256 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

The two projectiles are projected with the same initial velocities at the 15° and 30° with respect to the horizontal. The ratio of their ranges is 1:$x$. The value of $x$ is

**Options :**

695278871. $\sqrt{2}$

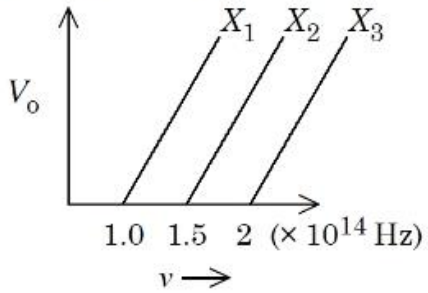
695278872. $\sqrt{3}$

695278873. $2\sqrt{3}$

695278874. $\frac{1}{\sqrt{2}}$

**Question Number : 32 Question Id : 695278257 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

The graph shows variation of stopping potential $V_o$ with the frequency $v$ of the incident radiation for three photosensitive metals $X_1$, $X_2$ and $X_3$. Which metal will give out electrons with greater kinetic energy, for the same wavelength of incident radiation?




**Options :**

695278875. $X_1$

695278876. $X_2$

695278877. $X_3$

695278878. All the metals will give out photo electrons with same kinetic energies.

**Question Number : 33 Question Id : 695278258 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

A slit of width $a$ is illuminated by light of wavelength λ. The linear separation between 1$^{st}$ and 3$^{rd}$ minima in the diffraction pattern produced on a screen placed at a distance $D$ from the slit system is _____.

**Options :**

695278879. $\frac{D\lambda}{a}$

695278880. $1.5\frac{D\lambda}{a}$

695278881. $2\frac{D\lambda}{a}$

695278882. $3\frac{D\lambda}{a}$

**Question Number : 34 Question Id : 695278259 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

A string $A$ of length 0.314 m and Young's modulus $2 \times 10^{10}$ N/m$^2$ is connected to another string $B$ of length and Young's modulus both twice of those of $A$. This series combination of strings is then suspended from a rigid support and its free end is fixed to a load of mass 0.8 kg. The net change in length of the combination is _____mm.
(radius of both the strings is 0.2 mm and acceleration due to gravity =10 m/s$^2$)
(Mass of both strings is to be neglected as compared to the mass of load)

**Options :**

695278883. 3

695278884. 2

695278885. 1.9

695278886. 1

**Question Number : 35 Question Id : 695278260 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

One gas of $n_1$ mole of molecules at temperature $T_1$, volume $V_1$, and pressure $P_1$, and another gas of $n_2$ mole of molecules at temperature $T_2$, volume $V_2$, and pressure $P_2$, are mixed resulting in pressure $P$ and volume $V$ of the mixture. The temperature of the mixture is _____.

**Options :**

695278887. $(T_1 + T_2)/2$

695278888. $T_1 T_2 PV/(T_2 P_1V_1 + T_1 P_2V_2)$

695278889. $(T_2 P_1V_1 + T_1 P_2V_2)/(T_1 T_2 PV)$

695278890. $|T_1 - T_2|/2$

**Question Number : 36 Question Id : 695278261 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

An ideal gas undergoes a process maintaining relation between pressure ($P$) and volume($V$) as $P = P_o\left(1+\left(\frac{V_o}{V}\right)^2\right)^{-1}$, where $P_o$ and $V_o$ are constants. If two samples $A$ and $B$ (two moles each) with initial volumes $V_o$ and $3V_o$ respectively undergo above mentioned process and attain same pressure, then the difference at the temperatures of these samples, $T_B - T_A$ is _____.

($R$ = gas constant)

**Options :**

695278891. $\frac{9P_oV_o}{8R}$

695278892. $\frac{11P_oV_o}{10R}$

695278893. $\frac{7P_oV_o}{6R}$

695278894. $\frac{13P_oV_o}{11R}$

**Question Number : 37 Question Id : 695278262 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

A voltmeter with internal resistance of $x\ \Omega$ can be used to measure upto 20 V. In order to increase its measuring range to 30 V, the required modification is to_____.

**Options :**

695278895. connect resistor of $\frac{x}{2}\Omega$, in series with voltmeter.

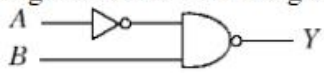
695278896. connect resistor of $\frac{x}{2}\Omega$, in parallel to voltmeter.

695278897. connect a resistor of $x\ \Omega$ in series with voltmeter.

695278898. connect resistor of $2x\ \Omega$ in parallel to voltmeter.

**Question Number : 38 Question Id : 695278263 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

Two 4 bits binary numbers, $A = 1101$ and $B = 1010$ are given in the inputs of a logic circuit shown in figure below. The output ($Y$) will be :




**Options :**

695278899. $Y = 1101$

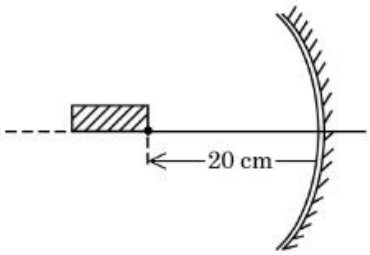
695278900. $Y = 0010$

695278901. $Y = 0111$

695278902. $Y = 1000$

**Question Number : 39 Question Id : 695278264 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

A rod of length 10 cm lies along the principle axis of a concave mirror of focal length 10 cm as shown in figure. The length of the image is _____ cm.




**Options :**

695278903. 2.5

695278904. 5

695278905. 7.5

695278906. 7

**Question Number : 40 Question Id : 695278265 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

A parallel plate air capacitor is connected to a battery. The plates are pulled apart at uniform speed $v$. If $x$ is the separation between the plates at any instant, then the time rate of change of electrostatic energy of the capacitor is proportional to $x^{\alpha}$, where $\alpha$ is _____.

**Options :**

695278907. $-2$

695278908. 1

695278909. $-1$

695278910. 2

**Question Number : 41 Question Id : 695278266 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

An insulated wire is wound so that it forms a flat coil with $N = 200$ turns. The radius of the innermost turn is $r_1 = 3$ cm, and of the outermost turn $r_2 = 6$ cm. If 20 mA current flows in it then the magnetic moment will be $\alpha \times 10^{-2}$ A.m$^2$. The value of $\alpha$ is_____.

**Options :**

695278911. 4.4

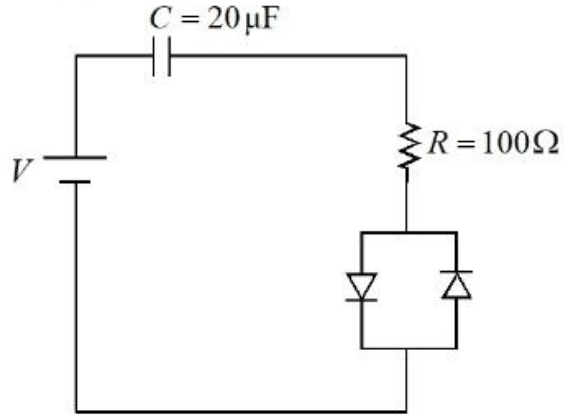
695278912. 2.64

695278913. 3.25

695278914. 1.2

**Question Number : 42 Question Id : 695278267 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

Consider a circuit consisting of a capacitor (20 μF), resistor (100 Ω) and two identical diodes as shown in figure. The resistance of diode under forward biasing condition is 10 Ω. The time constant of the circuit is $\alpha \times 10^{-3}$ s. The value of α is _____




**Options :**

695278915. 2.2

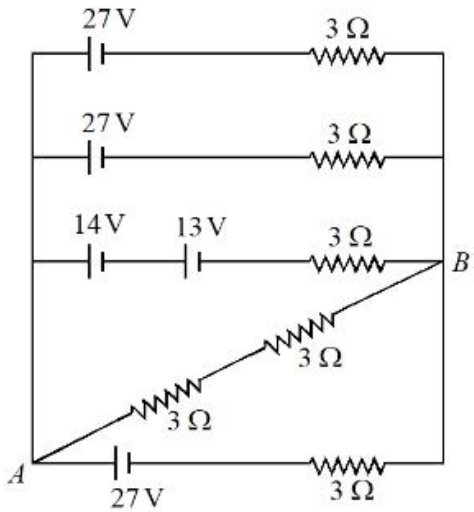
695278916. 2.0

695278917. 2.1

695278918. 2.4

**Question Number : 43 Question Id : 695278268 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

The voltage and the current between *A* and *B* points shown in the circuit are_____.




**Options :**

695278919. 24 V, 12 A

695278920. 24 V, 4 A

695278921. 18 V, 12 A

695278922. 27 V, 4 A

**Question Number : 44 Question Id : 695278269 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

A telescope with objective diameter $R$ is used to observe a distant star emitting light of wavelength 500 nm, at a resolution of $5 \times 10^{-7}$ radian. The value of $R$ is _____ cm.

**Options :**

695278923. 61

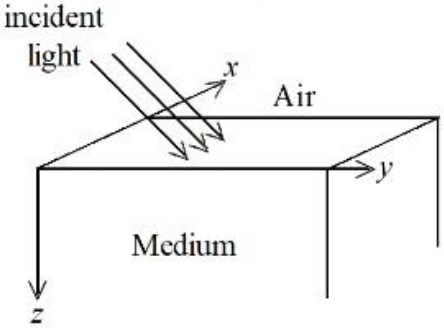
695278924. 122

695278925. 244

695278926. 305

**Question Number : 45 Question Id : 695278270 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

An unpolarized light is incident on the plane interface of air-dielectric medium shown in figure. If the incident angle is equal to Brewster angle, identify the expression representing reflected wave.




**Options :**

695278927. $(E_x\hat{i}+E_y\hat{j})\sin\left(kx-kz-\omega t\right)$

695278928. $(E_x\hat{i}+E_z\hat{k})\sin\left(kx+ky-\omega t\right)$

695278929. $(E_x\hat{j}+E_y\hat{k})\sin\left(ky+kz-\omega t\right)$

695278930. $(E_x\hat{i}+E_y\hat{j}+E_z\hat{k})\sin\left(kx+ky-kz-\omega t\right)$

## Physics Section B

| | |
|---|---|
| **Section Id :** | 69527822 |
| **Section Number :** | 4 |
| **Section type :** | Online |
| **Mandatory or Optional :** | Mandatory |
| **Number of Questions :** | 5 |
| **Number of Questions to be attempted :** | 5 |
| **Section Marks :** | 20 |
| **Maximum Instruction Time :** | 0 |
| **Sub-Section Number :** | 1 |
| **Sub-Section Id :** | 69527822 |
| **Question Shuffling Allowed :** | Yes |
| **Is Section Default? :** | No |

**Question Number : 46 Question Id : 695278271 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript**

A 1 kg block subjected to two simultaneous forces $\left(2\hat{i}+3\hat{j}+4\hat{k}\right)$ N and $\left(3\hat{i}-\hat{j}-2\hat{k}\right)$ N is moved a distance of 25 m along $\left(3\hat{i}-4\hat{j}\right)$ direction. The work done in this process is _____J.

**Response Type :** Numeric
**Evaluation Required For SA :** Yes
**Show Word Count :** Yes
**Answers Type :** Equal
**Text Areas :** PlainText
**Possible Answers :**
1

**Question Number : 47 Question Id : 695278272 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript**

The surface tension of a soap solution is $3.5 \times 10^{-2}$ N/m. The work required to increase the radius of a soap bubble from 1 cm to 2 cm is $\alpha \times 10^{-6}$ J. The value of $\alpha$ is _____.
$(\pi = 22/7)$

**Response Type :** Numeric
**Evaluation Required For SA :** Yes
**Show Word Count :** Yes
**Answers Type :** Equal
**Text Areas :** PlainText
**Possible Answers :**
1

**Question Number : 48 Question Id : 695278273 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript**

The velocity of a particle executing simple harmonic motion along $x$-axis is described as $v^2 = 50 - x^2$, where $x$ represents displacement. If the time period of motion is $\frac{x}{7}$ s, the value of $x$ is _____.

**Response Type :** Numeric
**Evaluation Required For SA :** Yes
**Show Word Count :** Yes
**Answers Type :** Equal
**Text Areas :** PlainText
**Possible Answers :**
1

**Question Number : 49 Question Id : 695278274 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript**

A body of mass 2 kg begins to move under the influence of time dependent force $\vec{F} = \left(2t\,\hat{i} + 6t^2\,\hat{j}\right)$ N, where $\hat{i}$ and $\hat{j}$ are unit vectors along $x$ and y-axis respectively. The power produced by the force at $t = 2$ s is ____W.

**Response Type :** Numeric
**Evaluation Required For SA :** Yes
**Show Word Count :** Yes
**Answers Type :** Equal
**Text Areas :** PlainText
**Possible Answers :**
1

**Question Number : 50 Question Id : 695278275 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript**

An inductor of 10 mH, capacitor of 0.1 μF and a resistor of 100 Ω are connected in series across an *a.c* power supply 220 V, 70 Hz. The power factor of the given circuit is 0.5. The difference in the inductive reactance and capacitance reactance is $\sqrt{3}\ \alpha\ \Omega$. The value of α is _____.

**Response Type :** Numeric
**Evaluation Required For SA :** Yes
**Show Word Count :** Yes
**Answers Type :** Equal
**Text Areas :** PlainText
**Possible Answers :**
1

# Chemistry Section A

| | |
|---|---|
| **Section Id :** | 69527823 |
| **Section Number :** | 5 |
| **Section type :** | Online |
| **Mandatory or Optional :** | Mandatory |
| **Number of Questions :** | 20 |
| **Number of Questions to be attempted :** | 20 |
| **Section Marks :** | 80 |
| **Maximum Instruction Time :** | 0 |
| **Sub-Section Number :** | 1 |
| **Sub-Section Id :** | 69527823 |
| **Question Shuffling Allowed :** | Yes |
| **Is Section Default? :** | No |

**Question Number : 51 Question Id : 695278276 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

Number of moles and number of molecules in 1.4187 L of $SO_2$ at STP respectively are

**Options :**

695278936. $0.1266; 3.812 \times 10^{22}$

695278937. $0.0633; 3.812 \times 10^{22}$

695278938. $0.1266; 7.6238 \times 10^{22}$

695278939. $0.0633; 7.6238 \times 10^{22}$

**Question Number : 52 Question Id : 695278277 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

What is the ratio of wave number of first line (lowest energy line) of Balmer series of H atomic spectrum to first line of its Brackett series?

**Options :**

695278940. 5:1

695278941. 5:0.81

695278942. 5:1.75